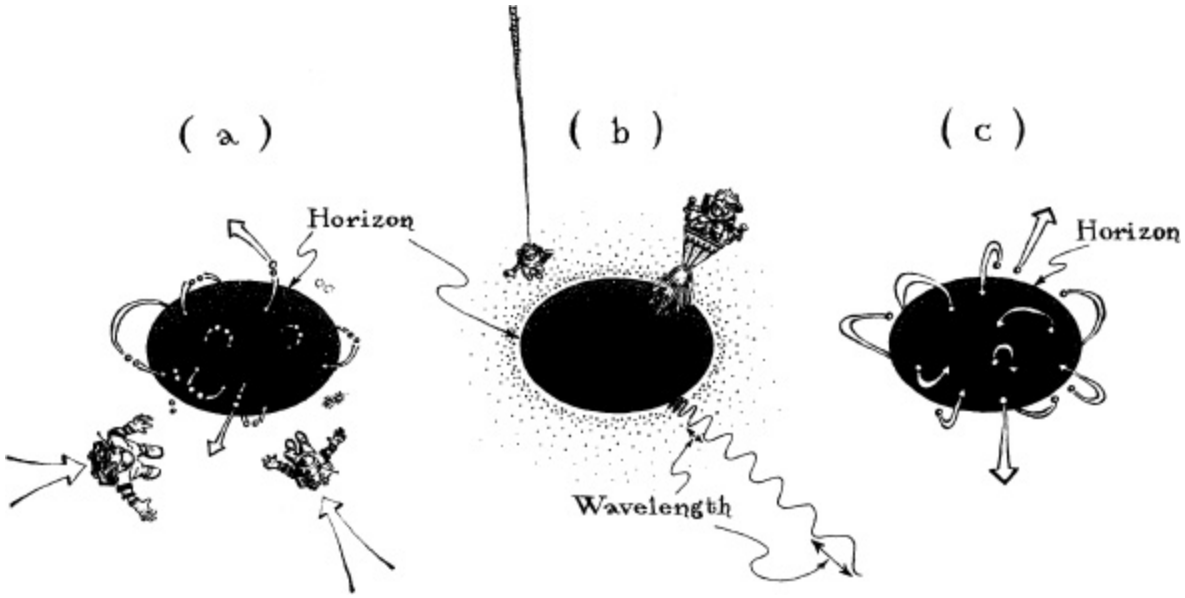
Figure 12.3b depicts a different viewpoint on the hole's vacuum fluctuations, the viewpoint of observers who reside just above the hole's horizon and are forever at rest relative to the horizon. To prevent themselves from being swallowed by the hole, such observers must accelerate hard, relative to falling observers—using a rocket engine or hanging by a rope. For this reason, these observers' viewpoint is called the "accelerated viewpoint." It is also the viewpoint of the "membrane paradigm" (Chapter 11).

Surprisingly, from the accelerated viewpoint, the vacuum fluctuations consist not of virtual particles flashing in and out of existence, but rather of real particles with positive energies and long lives; see Box 12.5. The real particles form a hot atmosphere around the hole, much like the atmosphere of the Sun. Associated with these real particles are real waves. As a particle moves upward through the atmosphere, gravity pulls on it, reducing its energy of motion; correspondingly, as a wave moves upward, it becomes gravitationally redshifted to longer and longer wavelengths (Figure 12.3b).



12.3 (a) Observers falling into a black hole (the two little men in space suits) see vacuum fluctuations near the hole's horizon to consist of pairs of virtual particles, (b) As viewed by observers just above the horizon and at rest relative to the horizon (the little man hanging by a rope and the little man blasting his rocket engine), the vacuum fluctuations consist of a hot atmosphere of real particles; this is the "accelerated viewpoint." (c) The atmosphere's particles, in the accelerated viewpoint, appear to be emitted by a hot, membrane-like horizon. They fly upward short distances, and most are then pulled back into the horizon. However, a few of the particles manage to escape the hole's grip and evaporate into outer space.